

Experiment No. 4

Student Name: Swapnil Gaurkar

PRN: 25070521504

Experiment Title:

Use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software / Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- CSS3
- JavaScript (ES6)

Experiment Program Code

Files Used:

- palindrome.html
- style5.css
- script6.js
- **palindrome.html**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Palindrome Checker</title>

  <link rel="stylesheet" href="style5.css">
</head>

<body>

  <!-- Student Details -->
  <div class="student-info">
    <h3>SWAPNIL GAURKAR</h3>
    <p>PRN: 25070521504</p>
    <p>Section: A</p>
  </div>

  <!-- Main Palindrome Card -->
  <div class="container">

    <h1>Palindrome Checker</h1>

    <input
      type="text"
      id="textInput"
      placeholder="Enter text"
      value="MADAM"
    >

    <button id="checkButton">
      Check Status
    </button>

    <div id="result" class="result success">
      ■ "MADAM" is a palindrome!
    </div>

  </div>

  <script src="script6.js"></script>

</body>
</html>
```

• style5.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
```

```

    min-height: 100vh;

    font-family: Arial, Helvetica, sans-serif;

    background:
        linear-gradient(
            120deg,
            #0e172c 0%,
            #121831 40%,
            #20184e 100%
        );

    color: white;

    display: flex;
    justify-content: center;
    align-items: center;

    position: relative;
}

/* =====
STUDENT INFO BOX
===== */

.student-info {
    position: absolute;

    top: 20px;
    right: 20px;

    width: 155px;

    background: rgba(28, 39, 64, 0.85);

    border: 1px solid rgba(117, 107, 255, 0.15);

    border-radius: 8px;

    padding: 14px 15px;

    text-align: right;

    box-shadow:
        0 5px 15px rgba(0, 0, 0, 0.18);
}

.student-info h3 {
    color: #c184ff;

    font-size: 13px;

    margin-bottom: 5px;
}

.student-info p {
    color: #aab2c8;

    font-size: 12px;

    line-height: 18px;
}

/* =====
MAIN CARD
===== */

.container {
    width: 360px;

    padding: 40px 36px 36px;

    background: #1b2940;

    border: 1px solid rgba(123, 137, 171, 0.25);

    border-radius: 14px;

    box-shadow:
        0 15px 40px rgba(0, 0, 0, 0.28);
}

/* =====
TITLE
===== */

.container h1 {
    text-align: center;

    font-size: 25px;

    margin-bottom: 24px;

    background: linear-gradient(

```

```

        90deg,
        #8d8cff,
        #c25cff
    );

    -webkit-background-clip: text;
    background-clip: text;

    color: transparent;
}

/* =====
INPUT
===== */

#textInput {
    width: 100%;

    height: 42px;

    padding: 0 14px;

    background: #131d31;

    color: #ffffff;

    border: 2px solid #43599e;

    border-radius: 7px;

    outline: none;

    font-size: 15px;

    transition: 0.2s;
}

#textInput:focus {
    border-color: #7764ff;

    box-shadow:
        0 0 8px rgba(119, 100, 255, 0.25);
}

#textInput::placeholder {
    color: #778198;
}

/* =====
BUTTON
===== */

#checkButton {
    width: 100%;

    height: 38px;

    margin-top: 14px;

    border: none;

    border-radius: 7px;

    background:
        linear-gradient(
            90deg,
            #5147e7,
            #8138ed
        );

    color: white;

    font-size: 14px;

    font-weight: bold;

    cursor: pointer;

    transition:
        transform 0.2s,
        box-shadow 0.2s;
}

#checkButton:hover {
    transform: translateY(-1px);

    box-shadow:
        0 5px 15px rgba(105, 67, 235, 0.35);
}

#checkButton:active {
    transform: translateY(0);
}

/* =====

```

```

    RESULT BOX
    ===== */

.result {
    width: 100%;

    margin-top: 22px;

    padding: 11px;

    border-radius: 6px;

    text-align: center;

    font-size: 15px;

    font-weight: bold;
}

/* Palindrome */

.success {
    color: #37d5ad;

    background: rgba(15, 112, 112, 0.25);

    border: 1px solid rgba(45, 207, 180, 0.25);
}

/* Not Palindrome */

.error {
    color: #ff7c91;

    background: rgba(145, 38, 65, 0.22);

    border: 1px solid rgba(255, 91, 120, 0.25);
}

/* =====
MOBILE RESPONSIVE
===== */

@media (max-width: 500px) {

    .container {
        width: 90%;
        padding: 35px 25px;
    }

    .student-info {
        top: 10px;
        right: 10px;
    }

    .container h1 {
        font-size: 22px;
    }
}

```

• script6.js

```

const input = document.getElementById("textInput");
const button = document.getElementById("checkButton");
const result = document.getElementById("result");

function checkPalindrome() {

    // Get user input
    const originalText = input.value.trim();

    // Empty input validation
    if (originalText === "") {

        result.textContent = "■■■ Please enter some text.";

        result.className = "result error";

        return;
    }

    // Convert text for palindrome checking
    // Example:
    // "Madam" -> "madam"
    // "Never odd or even" -> "neveroddoreven"

    const cleanedText = originalText
        .toLowerCase()
        .replace(/^[a-z0-9]/g, "");

    // Reverse text

```

```

const reversedText = cleanedText
    .split("")
    .reverse()
    .join("");

// Check palindrome
if (cleanedText === reversedText) {

    result.textContent =
        `■ "${originalText}" is a palindrome!`;

    result.className = "result success";

} else {

    result.textContent =
        `■ "${originalText}" is not a palindrome.`;

    result.className = "result error";

}

// Button click
button.addEventListener("click", checkPalindrome);

// Allow ENTER key
input.addEventListener("keydown", function(event) {

    if (event.key === "Enter") {
        checkPalindrome();
    }

});

```

Application Description

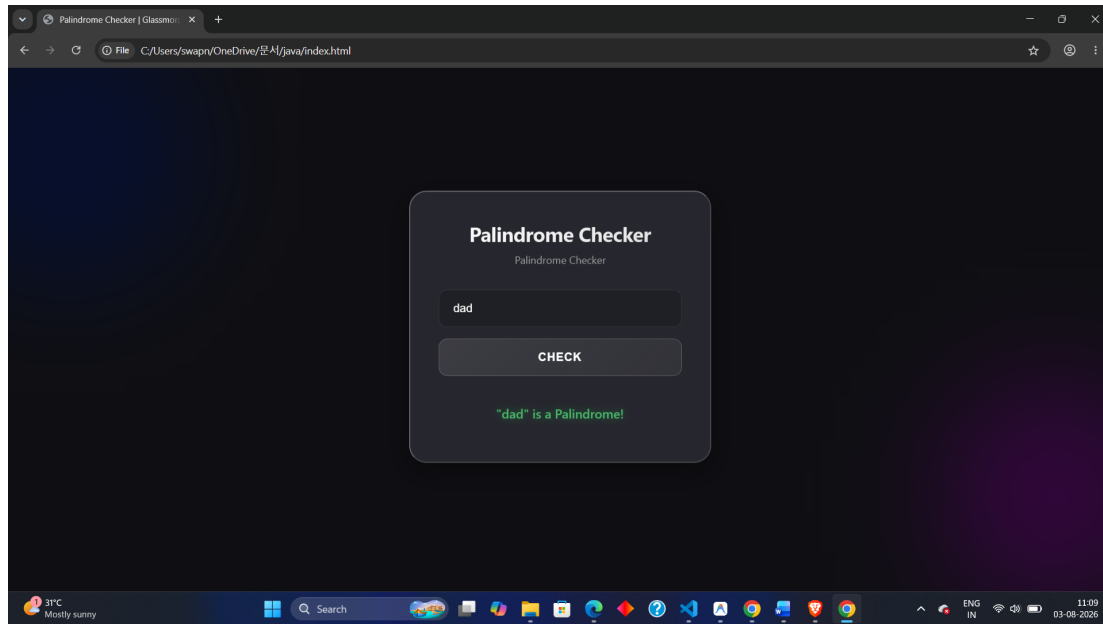
The Palindrome Checker accepts a string from the user and checks whether it reads the same forward and backward. The JavaScript program demonstrates function types, variable scope, closures, and exception handling using try-catch. The result is displayed dynamically on the webpage after the user clicks the Check button.

Features Implemented

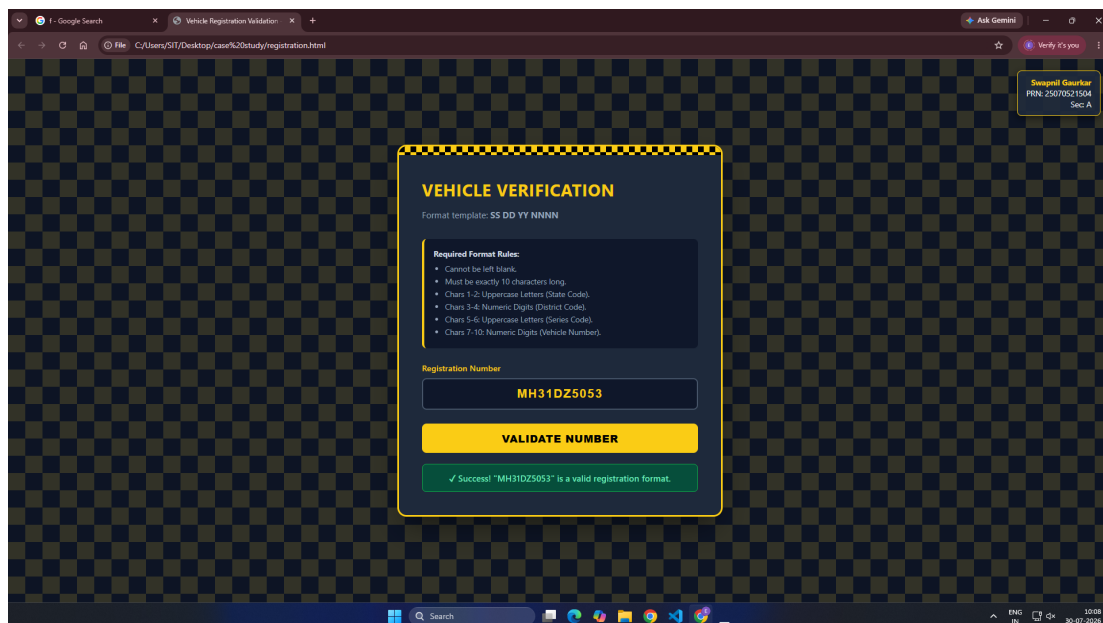
Feature	Description
Function Usage	Uses JavaScript functions to perform palindrome checking
Scope	Demonstrates local and function-level variable scope
Closures	Uses closure concepts within the JavaScript logic
Try-Catch	Handles errors and invalid situations using exception handling
String Processing	Processes the entered text to test palindrome condition
DOM Manipulation	Displays the result dynamically on the webpage
Event Handling	Check button triggers the palindrome checking logic

Output

Output 1 - Palindrome Checker



Output 2 - Vehicle Registration Validation



Output Summary

- The first output demonstrates the Palindrome Checker and shows that the entered string "dad" is identified as a palindrome.
- The second supplied output shows a Vehicle Registration Validation page successfully validating the registration number MH31DZ5053.
- Both supplied screenshots are included in the output section as provided.

Result / Conclusion

The Palindrome Checker was implemented successfully using HTML, CSS, and JavaScript. The experiment demonstrated function types, scope, closures, string processing, DOM manipulation, event handling, and try-catch based exception handling. The application successfully checks user-entered text and displays the result dynamically.